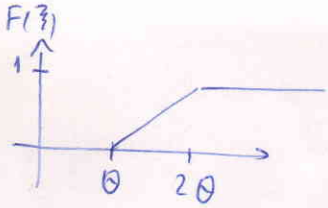
T5



а) выборка X_n, оценки параметра θ

1) найдем оценки параметра, ОММ

$$M\xi = \int_{\theta}^{2\theta} x \frac{1}{\theta} d\theta = \frac{x^2}{2\theta}\Big|_{\theta}^{2\theta} = \frac{3}{2}\theta$$

$$M\xi^2 = \int_{\theta}^{2\theta} x^2 \frac{1}{\theta} d\theta = \frac{x^3}{3\theta}\Big|_{0}^{2\theta} = \frac{7}{3}\theta^2$$

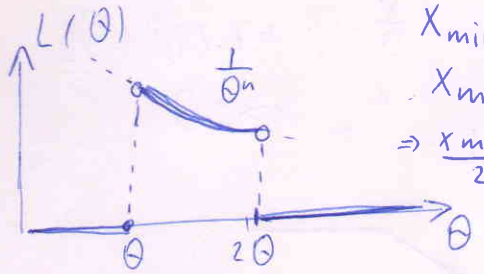
$$D\xi = M\xi^2 - (M\xi)^2 = \frac{7}{3}\theta^2 - \frac{9}{4}\theta^2 = \frac{1}{12}\theta^2$$

$$\bar{X} = \frac{3}{2}\theta \Rightarrow \tilde{\theta}_1 = \frac{2}{3}\bar{X}$$

2) ОМП

$$L(\theta) = \prod_{i=1}^{n} p(x_i, \theta) = \left(\frac{1}{\theta}\right)^n$$

$$\frac{\partial}{\partial\theta}(\ln L) = \frac{\partial}{\partial\theta}(-n\ln\theta) = \frac{-n}{\theta}$$



$X_{min} \geq \theta$

$X_{max} \leq 2\theta$

$\Rightarrow \frac{X_{max}}{2} \leq \theta \leq X_{min}$

$$\tilde{\theta}_2 = \frac{X_{max}}{2}$$

б) 1) $M\tilde{\theta}_1 = \frac{2}{3}M\xi = \theta$ несмещ

$$D\tilde{\theta}_1 = \frac{4}{9}D\bar{X} = \frac{4}{9n}D\xi = \frac{4}{9n}\frac{1}{12}\theta^2 = \frac{\theta^2}{27\cdot n} \xrightarrow[n\to\infty]{} 0$$ дост усл сост.

2) $M\tilde{\theta}_2 = \frac{1}{2}M X_{max}$ $\qquad X_{max} \sim \underbrace{(F(x))^n}_{\Psi(x)}$

$$\psi(x) = \Psi'(x) = n\left(\frac{x}{\theta}-1\right)^{n-1}\frac{1}{\theta}\{(\theta, 2\theta)\}$$

$$M X_{max} = \int_{\theta}^{2\theta} x\cdot n\left(\frac{x}{\theta}-1\right)^{n-1}\frac{1}{\theta}dx = \int_0^1 (t+1)\theta n t^{n-1}dt = \int_0^1 t^n\cdot\theta n\,dt + \int_0^1 t^{n-1}\theta n\,dt$$

$$= \frac{\theta n}{n+1} + \theta = \frac{2n+1}{n+1}\theta$$

$\Rightarrow M\tilde{\theta}_2 = \frac{n+1/2}{n+1}\theta$ смещ оценка

$$\tilde{\theta}_2' = \frac{n+1}{n+1/2}\cdot\frac{X_{max}}{2} = \frac{n+1}{2n+1}X_{max}, \quad M\tilde{\theta}_2' = \theta \text{ несмещ}$$

$$D\tilde{\theta}_2' = \left(\frac{n+1}{2n+1}\right)^2\frac{n\theta^2}{(n+2)(n+1)^2} = \frac{n}{(2n+1)^2(n+2)}\theta^2 \xrightarrow[n\to\infty]{} 0$$ дост усл сост

аналог. T1

c) $D\tilde{\Theta}_1 = \frac{\Theta^2}{27n}$

$D\tilde{\Theta}_2' = \frac{n\Theta^2}{(2n+1)^2(n+2)}$

при $n=1$ и $n \geq 3$ $\tilde{\Theta}_2'$ эфф $\tilde{\Theta}_1$

d) точный доверительный интервал

модель h

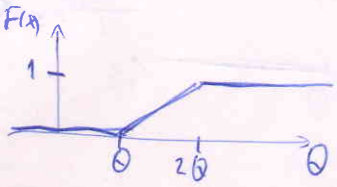
выборка $\overline{X_n}$ $\quad f(\vec{X_n}, h) \sim g(t)$

$$F(t) = \begin{cases} \frac{t}{\Theta} - 1 & t \geq \Theta \\ 0 & t < \Theta \end{cases}$$

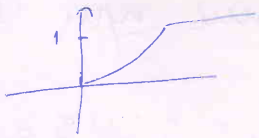
$$f(X_n, \Theta) = \frac{\max X_i}{\Theta} - 1$$

$$P(f < t) = P(X_{max} < \Theta t + \Theta) = (F(\Theta t + \Theta))^n$$

$$F(x) = \begin{cases} 0, & x < \Theta \\ \frac{x}{\Theta} - 1, & x \in [\Theta, 2\Theta] \\ 1, & x > 2\Theta \end{cases} \Rightarrow \begin{cases} 0, & t \leq 0 \\ t^n, & 0 < t \leq 1 \\ 1, & t > 1 \end{cases}$$



$$t_1 = q_{\frac{1-\beta}{2}} = q_{\frac{\alpha}{2}} = \sqrt[n]{\frac{\alpha}{2}} = \sqrt[n]{\frac{1-\beta}{2}}$$



$$t_2 = q_{\frac{1+\beta}{2}} = \sqrt[n]{\frac{1+\beta}{2}}$$

$$P\left(t_1 < \frac{X_{max}}{\Theta} - 1 < t_2\right) = \beta$$

$$P\left(\frac{X_{max}}{1 + \sqrt[n]{\frac{1+\beta}{2}}} < \Theta < \frac{X_{max}}{1 + \sqrt[n]{\frac{1-\beta}{2}}}\right) = \beta$$

если $\beta = 0{,}95$, то $\frac{X_{max}}{1+\sqrt[n]{0{,}975}} < \Theta < \frac{X_{max}}{1+\sqrt[n]{0{,}025}}$

дов. интервал;

e) асимптотический дов. инт.

ОММ:

$$g(\mathcal{L}_1) = \Theta^2 = \frac{2}{3}\mathcal{L}_1$$

$$\frac{\sqrt{n}(\hat{\Theta} - \Theta)}{\sqrt{\frac{4}{9}(\mathcal{L}_2 - \mathcal{L}_1^2)}} \rightsquigarrow N(0,1)$$

$t_1 = U_{\frac{1-\beta}{2}} = -1{,}96 \quad t_2 = U_{\frac{1+\beta}{2}} = 1{,}96$

$$-\frac{1{,}96\sqrt{\frac{4}{9}(\mathcal{L}_2 - \mathcal{L}_1^2)}}{\sqrt{n}} + \frac{2}{3}\mathcal{L}_1 < \Theta < \frac{1{,}96\sqrt{\frac{4}{9}(\mathcal{L}_2 - \mathcal{L}_1^2)}}{\sqrt{n}} + \frac{2}{3}\mathcal{L}_1$$

дов интервал